

YILUN ZHOU

CONTACT

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EDUCATION

Massachusetts Institute of Technology (MIT) *June 2019 - February 2023*
Doctor of Philosophy (Ph.D.) *GPA: 5.0 / 5.0*
Department of Electrical Engineering and Computer Science (EECS)
Advisor: Julie Shah
Thesis: Techniques for Interpretability and Transparency of Black-Box Models

Massachusetts Institute of Technology (MIT) *Sept 2017 - June 2019*
Master of Science (M.S.) *GPA: 5.0 / 5.0*
Department of Electrical Engineering and Computer Science (EECS)
Advisor: Julie Shah

Duke University *August 2013 - December 2016*
Bachelor of Science in Engineering (B.S.E.) *GPA: 3.98 / 4.0*
Department of Computer Science
Department of Electrical and Computer Engineering
Advisor: George Konidakis and Kris Hauser

RESEARCH

My research mainly aims to understand modern deep learning models, in particular large language models (LLMs). On the one hand, I assess the inner workings of these models by developing novel interpretability analyses. On the other hand, I curate novel benchmark tasks to assess diverse model behaviors, such their creativity, hallucination, reasoning, and safety. These two directions together lead to more fine-grained model insights, which allows for model improvement in a more dedicated and efficient manner.

WORK EXPERIENCE

Salesforce Research, Senior Research Scientist *February 2024 - Present*

- Part of the Interactive AI Group, working on trust and safety-related research and product for large language models.
- Started and contributed to xAlign, a library for LLM alignment with human preference, which is currently being deployed as an internal LLM finetuning toolkit.
- Mentored two summer research internship projects with under-review publications.
- Led and involved in three research projects with under-review publications.

Amazon, Applied Scientist II *January 2023 - February 2024*

- Part of the advertisement technology team, working on scheduling and delivering video advertisements on Amazon and 3rd party platforms.
- Conducted research prototyping on constraint-aware ad scheduling.
- Worked with the engineering team to deploy research prototype into production.

Microsoft Research, Research Intern

May 2022 - August 2022

NVIDIA, Research Intern

May 2021 - September 2021

Facebook AI, Research Intern

May 2020 - September 2020

TALKS

- **Math AI Workshop @ ICML 2024.** How and Where Do LLMs Fail on Math? A Multi-Faceted Evaluation of LLM Math Reasoning Skills. July 2024.
- **Amazon.** Algorithmic and Societal Characteristics of Model Explanations. August 2023.
- **Brown University.** Correctness and Understandability of Model Interpretability Methods. June 2022.
- **University of Michigan** (Guest Lecture for EECS 692: Advanced AI). Methods and Evaluations for *Post Hoc* Model-Agnostic Local Explanations. March 2022.
- **Meta AI.** The Missing User Manual for Model Interpretability Methods: Evaluation, Comprehension and Integration. March 2022.
- **Future of Privacy Forum.** Model Explanations: Hopes, Setbacks and Paths Forward. February 2022.

SERVICE

Conference reviewer: ICML, NeurIPS, ICLR, AISTATS, AAAI, IJCAI, ACL, NAACL, EMNLP, CoRL, IROS

Journal reviewer: T-ASE, T-Cyb, IJHCI

Student volunteer: AISTATS 2021

Outstanding reviewer recognition: ICML 2022

PUBLICATIONS

Reverse chronological order. *Equal contribution.

J: journal. C: conference. W: workshop. P: pre-print. Highlighted work.

P5 Austin Xu, **Yilun Zhou**, Xuan-Phi Nguyen, Caiming Xiong, Shafiq Joty. J4R: Learning to Judge with Equivalent Initial State Group Relative Policy Optimization. *arXiv preprint:2505.13346*.

C16 **Yilun Zhou***, Austin Xu*, Peifeng Wang, Caiming Xiong, Shafiq Joty. Evaluating Judges as Evaluators: The JETTS Benchmark of LLM-as-Judges as Test-Time Scaling Evaluators. *International Conference on Machine Learning (ICML)*, 2025.

C15 Pranav Narayanan Venkit, Philippe Laban, **Yilun Zhou**, Yixin Mao, Chien-Sheng Wu. Search Engines in the AI Era: A Qualitative Understanding to the False Promise of Factual and Verifiable Source-Cited Responses in LLM-based Search. *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, June 2025.

C14 Fan Yin, Philippe Laban, Xiangyu Peng, **Yilun Zhou**, Yixin Mao, Vaibhav Vats, Linnea Ross, Divyansh Agarwal, Caiming Xiong, Chien-Sheng Wu. BingoGuard: LLM Content Moderation Tools with Risk Levels. *International Conference on Learning Representations (ICLR)*, April 2025.

P5 Peifeng Wang*, Austin Xu*, **Yilun Zhou**, Caiming Xiong, Shafiq Joty. Direct Judgement Preference Optimization. *arXiv preprint:2409.14664*.

P4 **Yilun Zhou**, Caiming Xiong, Silvio Savarese and Chien-Sheng Wu. Shared Imagination: LLMs Hallucinate Alike. *arXiv preprint:2407.16604*.

- P3 Rithesh Murthy*, Liangwei Yang*, Juntao Tan*, Tulika Manoj Awalgaoonkar*, **Yilun Zhou**, Shelby Heinicke, Sachin Desai, Jason Wu, Ran Xu, Sarah Tan, Jianguo Zhang, Zhiwei Liu, Shirley Kokane, Zuxin Liu, Ming Zhu, Huan Wang, Caiming Xiong, Silvio Savarese. MobileAIBench: Benchmarking LLMs and LMMs for On-Device Use Cases. *arXiv preprint:2406.10290*.
- P2 Daking Rai, **Yilun Zhou**, Shi Feng, Abulhair Saparov and Ziyu Yao. A Practical Review of Mechanistic Interpretability for Transformer-Based Language Models. *arXiv preprint:2407.02646*.
- C13 Yujun Mao, Yoon Kim and **Yilun Zhou**. CHAMP: A Competition-level Dataset for Fine-Grained Analyses of LLMs' Mathematical Reasoning Capabilities. *Annual Meeting of the Association for Computational Linguistics (ACL) Findings, August 2024*
- P1 Shiyuan Huang, Siddarth Mamidanna, Shreedhar Jangam, **Yilun Zhou** and Leilani H. Gilpin. Can Large Language Models Explain Themselves? A Study of LLM-Generated Self-Explanations. *arXiv preprint:2310.11207*.
- W6 Shawn Im, Jacob Andreas and **Yilun Zhou**. Evaluating the Utility of Model Explanations for Model Development. *Under Review*.
- C12 **Yilun Zhou**. Iterative Partial Fulfillment of Counterfactual Explanations: Benefits and Risks. *AAAI/ACM Conference on Artificial Intelligence, Ethics, and Society (AIES)*, August 2023.
- C11 Daking Rai, Bailin Wang, **Yilun Zhou** and Ziyu Yao. Improving Generalization in Language Model-based Text-to-SQL Semantic Parsing: Two Simple Semantic Boundary-Based Techniques. *Annual Meeting of the Association for Computational Linguistics (ACL)*, July 2023.
- C10 **Yilun Zhou** and Julie Shah. The Solvability of Interpretability Evaluation Metrics. *Conference of the European Chapter of the Association for Computational Linguistics (EACL) Findings*, May 2023.
- W5 Daking Rai, **Yilun Zhou**, Bailin Wang and Ziyu Yao. Explaining Large Language Model-Based Neural Semantic Parsers. *AAAI Student Abstract and Poster Program*, February 2023.
- C9 **Yilun Zhou**, Marco Tulio Ribeiro, and Julie Shah. EXSUM: From Local Explanations to Model Understanding. *Conference of the North American Chapter of the Association for Computational Linguistics – Human Language Technology (NAACL-HLT)*, July 2022.
- W4 Yiming Zheng, Serena Booth, Julie Shah, and **Yilun Zhou**. The Irrationality of Neural Rationale Models. *NAACL Workshop on Trustworthy Natural Language Processing (TrustNLP)*, July 2022.
- C8 Ganesh Ghalme*, Vineet Nair*, Vishakha Patil*, and **Yilun Zhou***. Long-Term Resource Allocation Fairness in Average Markov Decision Process (AMDP) Environment. *International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS)*, May 2022.
- J3 Mycal Tucker, **Yilun Zhou**, and Julie Shah. Latent Space Alignment Using Adversarially Guided Self-Play. *International Journal of Human-Computer Interaction (IJHCI)*, February 2022.
- C7 **Yilun Zhou**, Serena Booth, Marco Tulio Ribeiro, and Julie Shah. Do Feature Attributions Correctly Attribute Features? *AAAI Conference on Artificial Intelligence (AAAI)*, February 2022.
- C6 **Yilun Zhou**, Serena Booth, Nadia Figueroa, and Julie Shah. ROCUS: Robot Controller Understanding via Sampling. *Conference on Robot Learning (CoRL)*, November 2021.
- C5 **Yilun Zhou**, Adithya Renduchintala, Xian Li, Sida Wang, Yashar Mehdad, and Asish Ghoshal. Towards Understanding the Behaviors of Optimal Deep Active Learning Algorithms. *Artificial Intelligence and Statistics (AISTATS)*, April 2021.
- C4 Serena Booth*, **Yilun Zhou***, Ankit Shah, and Julie Shah. BAYES-TREX: a Bayesian Sampling Approach to Model Transparency by Example. *AAAI Conference on Artificial Intelligence (AAAI)*, February 2021.

- W3 Serena Booth*, Ankit Shah*, **Yilun Zhou***, and Julie Shah. Sampling Prediction-Matching Examples in Neural Networks: A Probabilistic Programming Approach. *AAAI Conference on Artificial Intelligence (AAAI) Workshop on Statistical Relational AI*, February 2020.
- W2 **Yilun Zhou**, Julie Shah, and Steven Schockaert. Learning Household Task Knowledge from WikiHow Descriptions. *International Joint Conference on Artificial Intelligence (IJCAI) Workshop on Semantic Deep Learning*, August 2019.
- C3 **Yilun Zhou**, Steven Schockaert, and Julie Shah. Predicting ConceptNet Path Quality Using Crowdsourced Assessments of Naturalness. *The Web Conference (WWW)*, May 2019.
- J2 **Yilun Zhou**, Benjamin Burchfiel, and George Konidaris. Representing, Learning, and Controlling Complex Object Interactions. *Autonomous Robots (AuRo)*, April 2018.
- W1 **Yilun Zhou** and Kris Hauser. 6DOF Grasp Planning by Optimizing a Deep Learning Scoring Function. *Robotics: Science and Systems (RSS) Workshop on Revisiting Contact - Turning a Problem into a Solution*, July 2017.
- C2 **Yilun Zhou** and Kris Hauser. Incorporating Side-Channel Information into Convolutional Neural Networks for Robotic Tasks. *IEEE International Conference on Robotics and Automation (ICRA)*, May 2017.
- J1 Kris Hauser and **Yilun Zhou**. Asymptotically Optimal Planning by Feasible Kinodynamic Planning in a State-Cost Space. *IEEE Transactions on Robotics (TRO)*, December 2016.
- C1 **Yilun Zhou** and George Konidaris. Representing and Learning Complex Object Interactions. *Robotics: Science and Systems (RSS)*, June 2016.